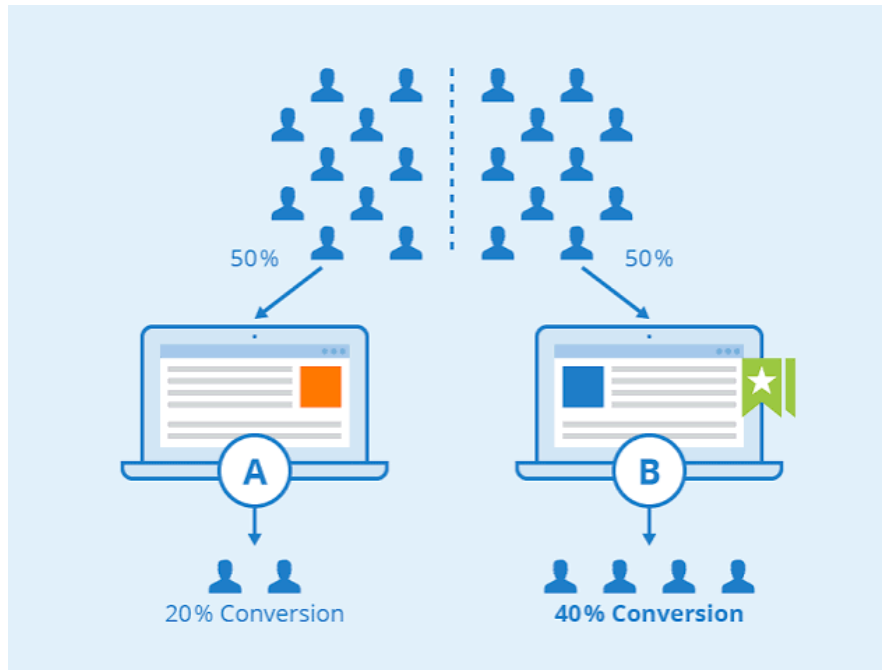


A/B Testing



A/B Testing is a statistical experiment where we compare two versions of something to determine which performs better.

- Version A = Original (Control)
- Version B = New Version (Treatment)

Simple Example

Netflix changes

Old Button

- Watch Now



New Button

- Start Watching

Question

Did more users click?

Only statistics can answer.

Why Companies Use A/B Testing

Companies don't rely on assumptions.

Instead of saying - **"I think red is better"**

They say - **"Let's collect data."**

Examples

Google	Search page changes
Amazon	Product recommendation
Facebook	Feed ranking
YouTube	Thumbnail testing
Instagram	Story layouts
Spotify	Playlist recommendation
Netflix	Movie posters
Uber	Surge pricing
Swiggy	Coupon banners
Zomato	Food recommendation

Real-Life Examples

Current Button Buy Now New Button Purchase Goal • Increase sales	Current Email Subject 50% Offer New Email Exclusive Offer Just For You Measure • Open Rate
Current Landing Page Blue Theme New Landing Page Green Theme Measure • Signups	Current Checkout 5 Steps New Checkout 2 Steps Measure • Completed Purchases

Basic Terminologies

Control Group

- Existing Version
- Version A

Treatment Group

- New Version
- Version B

Random Assignment

- Users should be assigned randomly.
- Never manually.

Conversion

- Desired action.



Conversion Rate

- Formula

$$\text{Conversion Rate} = \frac{\text{Conversions}}{\text{Visitors}}$$

- Example

1000 Visitors

80 Purchases

Conversion Rate

80/1000

=8%



Hypothesis

Null Hypothesis (H_0)

No difference.

Example

- Blue button and Green button have the same conversion rate.

Alternative Hypothesis (H_1)

There is a difference.

Green button performs better.

Types of A/B Tests

One-sided Test

Question

Is B better than A?

Example

Green button should increase sales.

Two-sided Test

Question

Is B different from A?

Could be better

Could be worse

Scenario

An e-commerce company wants to increase the number of customers who purchase products.

Currently, the website has a **blue "Buy Now" button**.

The marketing team believes changing it to a **green button** will increase purchases.

Instead of guessing, they conduct an **A/B Test**.

Step 1: Create Two Groups

The company randomly divides visitors into two equal groups.

Group	Button Color	Visitors
A (Control)	Blue	5000
B (Treatment)	Green	5000

Random assignment ensures both groups are similar.

Step 2: Collect Data

After one week, they collect the following data.

Group	Visitors	Purchases
A	5000	400
B	5000	470

Question:

Is the green button actually better, or is this difference due to random chance?

This is what A/B Testing answers.

Step 3: Calculate Conversion Rate

$$\text{Conversion Rate} = \frac{\text{Conversions}}{\text{Visitors}}$$

Version A

Visitors = 5000

Purchases = 400

$$\text{CRA} = \frac{400}{5000} = 0.08$$

Multiply by 100

$$\text{CRA} = 8\%$$

Version B

Visitors = 5000

Purchases = 470

$$\text{CRA} = \frac{470}{5000} = 0.094$$

Multiply by 100

$$\text{CRB} = 9.4\%$$

Current Observation

Version	Conversion Rate
A	8%
B	9.4%

Version B looks better.

But...

Is this difference statistically significant?

We don't know yet.

Step 4: State the Hypotheses

Null Hypothesis (H_0)

There is **no difference** between the two buttons.

$$H_0: p_A = p_B$$

Alternative Hypothesis (H_1)

There **is a difference** between the two buttons.

$$H_1: p_A \neq p_B$$

Since we only want to know if they are different, this is a **two-tailed test**.

Step 5: Significance Level

Choose

$\alpha=0.05$

Meaning

We are willing to accept a **5% chance of making a Type I Error (False Positive)**.

Decision rule

If

$p < 0.05$

Reject H_0

Otherwise

Fail to reject H_0

Step 6: Calculate the Pooled Proportion

Why do we calculate the pooled proportion?

Under the null hypothesis, we assume **both versions have the same true conversion rate**.

Instead of using two separate conversion rates, we estimate **one common conversion rate** using all the data.

Formula

$$p = \frac{x_1 + x_2}{n_1 + n_2}$$

Where

- **x1** = purchases in Group A = 400
- **x2** = purchases in Group B = 470
- **n1** = visitors in Group A = 5000
- **n2** = visitors in Group B = 5000

Substitute the values

$$p = \frac{400 + 470}{5000 + 5000} = \frac{870}{10000} = 0.087$$

The pooled conversion rate is **p=0.087** or **8.7%**

Step 7: Calculate Standard Error

Why Standard Error?

The observed difference (1.4%) could simply occur due to random sampling.

Standard Error measures the **expected variability** in the difference between two sample proportions if the null hypothesis is true.

Formula

$$\begin{aligned} \text{SE} &= \sqrt{p(1 - p)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} \\ &= \sqrt{0.087(1 - 0.087)\left(\frac{1}{5000} + \frac{1}{5000}\right)} \end{aligned}$$

First,

$$1 - 0.087 = 0.913$$

Multiply

$$0.087 \times 0.913 = 0.079431$$

Next,

$$\frac{1}{5000} = 0.0002$$

Now multiply

$$0.079431 \times 0.00040 = 0.0000317724$$

Take the square root

$$\sqrt{0.0000317724} = 0.00564$$

$$SE = 0.00564$$

Step 8: Calculate the Z Statistic

Formula

$$Z = \frac{pB - pA}{SE}$$

$$Z = \frac{0.094 - 0.08}{0.00564} = \frac{0.014}{0.00564} = 2.48$$

Interpretation: The observed difference is 2.48 standard errors away from the value expected under the null hypothesis.

Step 9: Find the p-value

From the standard normal (Z) table:

From the standard normal (Z) table:

- Area to the left of $Z=2.48 \approx 0.9934$
- Right-tail area = $1-0.9934=0.0066$

Since this is a **two-tailed test**:

$$p = 2 \times 0.0066 = 0.0132$$

So,

$$p \approx 0.013$$

Step 10: Make the Decision

Compare:

- p-value = **0.013**
- Significance level = **0.05**

Since

$$0.013 < 0.05$$

Reject the null hypothesis.

Step 11: Business Interpretation

Statistically,

The green button performs significantly better.

Business-wise:

- Blue button conversion = **8.0%**
- Green button conversion = **9.4%**
- Absolute improvement = **1.4 percentage points**

If the website receives **100,000 visitors per month**:

Old design:

$$100000 \times 8\% = 8000$$

New design:

$$100000 \times 9.4\% = 9400$$

Additional purchases:

$$9400 - 8000 = 1400$$

If the average profit per purchase is **₹500**:

Additional monthly profit:

$$1400 \times 500 = ₹700,000$$

This demonstrates that even a **1.4% increase in conversion rate** can have a **substantial business impact**.

Summary Table

Step	Result
Visitors (A)	5000
Visitors (B)	5000
Purchases (A)	400
Purchases (B)	470
Conversion Rate (A)	8.0%
Conversion Rate (B)	9.4%
Difference	1.4%
Pooled Proportion	0.087

Standard Error	0.00564
Z Statistic	2.48
p-value	0.013
Decision	Reject H_0
Winner	Version B (Green Button)